Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Cal-Denier Dairy LLC

Physical address of dairy:

Number and Street	City	County	Zip Code
10715 Arno RD	Galt	Sacramento	95632

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 08/01/1981

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X134-X280-X050-XXXX X150-X072-X003-XXXX

B. OPERATORS

Denier, Fred A

Operator name: Denier, Fred A

Telephone no.: (209) 745-4982 Landline, (209) 483-0318 Cellular

Mailing Address Number and Street	City	State	Zip Code
10715 Arno RD	Galt	CA	95632

Denier, Richard J.

Operator name: Denier, Richard J.

Telephone no.: (209) 745-4982 Landline, Cellular

Mailing Address Number and Street	City	State	Zip Code
10715 Arno RD	Galt	CA	95632

This operator is responsible for paying permit fees.

C. OWNERS

Denier, Fred A

Legal owner name: Denier, Fred A

Telephone no.: (209) 745-4982 Landline, (209) 483-0318 Cellular

Mailing Address Number and Street	City	State	Zip Code
10715 Arno RD	Galt	CA	95632

Denier, Richard J.

Legal owner name: Denier, Richard J.

Telephone no.: (209) 745-4982

Landline	Cellular
(209) 745-4982	

Mailing Address Number and Street	City	State	Zip Code
10715 Arno RD	Galt	CA	95632

This owner is responsible for paying permit fees.

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	0	110	0	0
Number under roof	850	100	180	0	0	0
Maximum number	850	100	180	110	0	0
Average number	850	100	180	110	0	0
Avg live weight (lbs)	1,400	1,450	1,000	750		

Predominant milk cow breed: Holstein

Average milk production: 85 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 27,711.51 *tons per reporting period*

Total nitrogen from manure: 350,408.45 *lbs per reporting period*

After ammonia losses (30% loss applied): 245,285.92 *lbs per reporting period*

Total phosphorus from manure: 58,924.91 *lbs per reporting period*

Total potassium from manure: 171,159.59 *lbs per reporting period*

Total salt from manure: 423,217.50 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 3,456,000 *gallons*

Total nitrogen generated: 18,169.40 *lbs*

Total phosphorus generated: 1,152.17 *lbs*

Total potassium generated: 23,086.68 *lbs*

Total salt generated: 234,183.40 *lbs*

	0	*gallons applied*
+	3,456,000	*gallons exported*
-	0	*gallons imported*
=	3,456,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Dairy/Dom. Well	Ground water
Marengo Irr. Well	Ground water
Tank Well	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

Date	Material type	Quantity	Reporting basis	Moisture (%)	Density (lbs/cu ft)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/17/2023	Corral solids	900.00 *ton*	Dry-weight	11.9		20,800.00	3,720.00	11,000.00		28.30

Date	Material type	Quantity	Kjeldahl-N (mg/L)	Ammonium-N (mg/L)	Ammonia-N (mg/L)	Nitrate-N (mg/L)	P (mg/L)	K (mg/L)	EC (µmhos/cm)	TDS (mg/L)
02/28/2023	Process wastewater	1,728,000.00 *gal*	599.00	398.00	0.00	0.00	54.00	612.00		6,380
09/30/2023	Process wastewater	1,728,000.00 *gal*	661.00	381.00	0.00	0.00	25.90	989.00		9,860

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Dry manure	32,984.64	5,899.18	17,443.80	448,781.40
Process wastewater	18,169.40	1,152.17	23,086.68	234,183.40
Total exports for all materials	51,154.04	7,051.35	40,530.48	682,964.80

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Field #D2	35	35	0	none	X134-X280-X050-XXXX
Field #D3	25	25	0	none	X134-X280-X050-XXXX
Field #D4	40	40	0	none	X134-X280-X050-XXXX
Field #D5	15	15	0	none	X134-X280-X050-XXXX
Field #D6	8	8	0	none	X134-X280-X050-XXXX
Field #M1	70	70	2	manure	X150-X072-X003-XXXX
Field #M2	30	30	1	manure	X150-X072-X003-XXXX
Field #M3	30	30	1	manure	X150-X072-X003-XXXX
Totals for areas that were used for application	130	130	4		
Totals for areas that were not used for application	123	123	0		
Land application area totals	253	253	4		

B. CROPS AND HARVESTS

Field #M1

Field name: Field #M1

06/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 70 Plant date: 06/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/03/2023	875.00 *ton*	Dry-weight		70.7	11,100.00	2,120.00	8,340.00		5.30
10/03/2023	875.00 *ton*	Dry-weight		68.9	11,700.00	2,130.00	7,660.00		6.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	25.00	172.28	32.09	120.65	885.83

Field #M2

Field name: Field #M2

Field #M2

06/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 30 Plant date: 06/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/04/2023	750.00 *ton*	Dry-weight		69.5	12,500.00	2,180.00	8,170.00		6.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	190.00	37.50	165.00	0.00
Total actual harvest content	25.00	190.63	33.25	124.59	1,037.00

Field #M3

Field name: Field #M3

06/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 30 Plant date: 06/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/04/2023	750.00 *ton*	Dry-weight		70.8	12,100.00	2,350.00	8,420.00		6.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	175.00	37.50	165.00	0.00
Total actual harvest content	25.00	176.66	34.31	122.93	949.00

NUTRIENT BUDGET

A. LAND APPLICATIONS

Field #M1 - 06/01/2023: Corn, silage

Field name: Field #M1

Crop: Corn, silage

Plant date: 06/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/25/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Application	Corral solids	111.06	28.56	179.82	0.00	950.00 *ton*
Application event totals		111.06	28.56	179.82	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Marengo Irr. Well	Ground water	9.28	0.00	0.00	0.00	7,788,000.00 *gal*
Application event totals		9.28	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Marengo Irr. Well	Ground water	8.89	0.00	0.00	0.00	7,458,000.00 *gal*
Application event totals		8.89	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Marengo Irr. Well	Ground water	8.42	0.00	0.00	0.00	7,062,000.00 *gal*
Application event totals		8.42	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Marengo Irr. Well	Ground water	8.42	0.00	0.00	0.00	7,062,000.00 *gal*
Application event totals		8.42	0.00	0.00	0.00	

Field #M1 - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Marengo Irr. Well	Ground water	8.42	0.00	0.00	0.00	7,062,000.00 *gal*
Application event totals		8.42	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Marengo Irr. Well	Ground water	8.42	0.00	0.00	0.00	7,062,000.00 *gal*
Application event totals		8.42	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Marengo Irr. Well	Ground water	8.42	0.00	0.00	0.00	7,062,000.00 *gal*
Application event totals		8.42	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Marengo Irr. Well	Ground water	8.42	0.00	0.00	0.00	7,062,000.00 *gal*
Application event totals		8.42	0.00	0.00	0.00	

Field #M2 - 06/01/2023: Corn, silage

Field name: Field #M2

Crop: Corn, silage

Plant date: 06/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/25/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Application	Corral solids	122.76	31.57	198.75	0.00	450.00 *ton*
Application event totals		122.76	31.57	198.75	0.00	

Field #M2 - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Marengo Irr. Well	Ground water	10.65	0.00	0.00	0.00	3,828,000.00 *gal*
Application event totals		10.65	0.00	0.00	0.00	

06/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Marengo Irr. Well	Ground water	9.73	0.00	0.00	0.00	3,498,000.00 *gal*
Application event totals		9.73	0.00	0.00	0.00	

07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Marengo Irr. Well	Ground water	8.63	0.00	0.00	0.00	3,102,000.00 *gal*
Application event totals		8.63	0.00	0.00	0.00	

07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Marengo Irr. Well	Ground water	8.63	0.00	0.00	0.00	3,102,000.00 *gal*
Application event totals		8.63	0.00	0.00	0.00	

08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Marengo Irr. Well	Ground water	8.63	0.00	0.00	0.00	3,102,000.00 *gal*
Application event totals		8.63	0.00	0.00	0.00	

08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Marengo Irr. Well	Ground water	8.63	0.00	0.00	0.00	3,102,000.00 *gal*
Application event totals		8.63	0.00	0.00	0.00	

Field #M2 - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Marengo Irr. Well	Ground water	8.63	0.00	0.00	0.00	3,102,000.00 *gal*
Application event totals		8.63	0.00	0.00	0.00	

09/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Marengo Irr. Well	Ground water	8.63	0.00	0.00	0.00	3,102,000.00 *gal*
Application event totals		8.63	0.00	0.00	0.00	

Field #M3 - 06/01/2023: Corn, silage

Field name: Field #M3

Crop: Corn, silage

Plant date: 06/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/25/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Application	Corral solids	122.76	31.57	198.75	0.00	450.00 *ton*
Application event totals		122.76	31.57	198.75	0.00	

06/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Marengo Irr. Well	Ground water	10.65	0.00	0.00	0.00	3,828,000.00 *gal*
Application event totals		10.65	0.00	0.00	0.00	

06/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Marengo Irr. Well	Ground water	9.73	0.00	0.00	0.00	3,498,000.00 *gal*
Application event totals		9.73	0.00	0.00	0.00	

Field #M3 - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Marengo Irr. Well	Ground water	8.63	0.00	0.00	0.00	3,102,000.00 *gal*
Application event totals		8.63	0.00	0.00	0.00	

07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Marengo Irr. Well	Ground water	8.63	0.00	0.00	0.00	3,102,000.00 *gal*
Application event totals		8.63	0.00	0.00	0.00	

08/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Marengo Irr. Well	Ground water	8.63	0.00	0.00	0.00	3,102,000.00 *gal*
Application event totals		8.63	0.00	0.00	0.00	

08/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Marengo Irr. Well	Ground water	8.63	0.00	0.00	0.00	3,102,000.00 *gal*
Application event totals		8.63	0.00	0.00	0.00	

09/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Marengo Irr. Well	Ground water	8.63	0.00	0.00	0.00	3,102,000.00 *gal*
Application event totals		8.63	0.00	0.00	0.00	

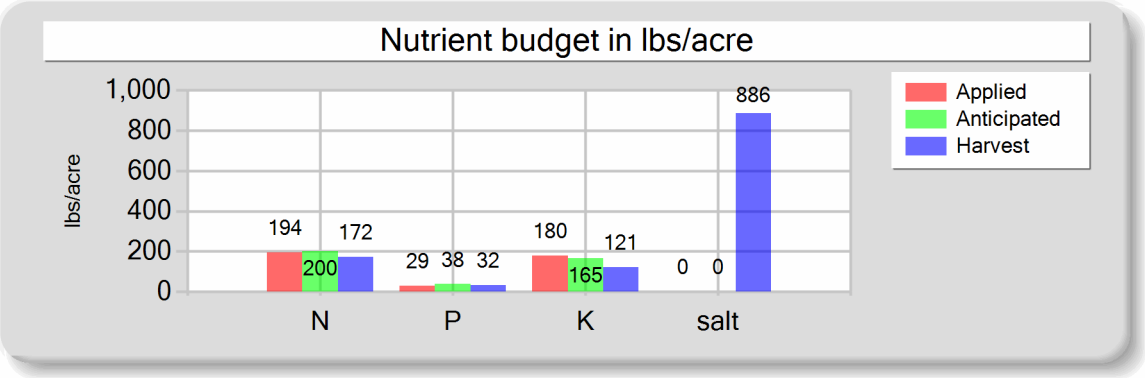
09/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Marengo Irr. Well	Ground water	8.63	0.00	0.00	0.00	3,102,000.00 *gal*
Application event totals		8.63	0.00	0.00	0.00	

B. NUTRIENT BUDGET

Field #M1 - 06/01/2023: Corn, silage

Field name: Field #M1 Crop: Corn, silage Plant date: 06/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	111.06	28.56	179.82	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	68.69	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	193.75	28.56	179.82	0.00
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	172.28	32.09	120.65	885.83
Nutrient balance	21.48	-3.53	59.17	-885.83
Applied to removed ratio	1.12	0.89	1.49	0.00

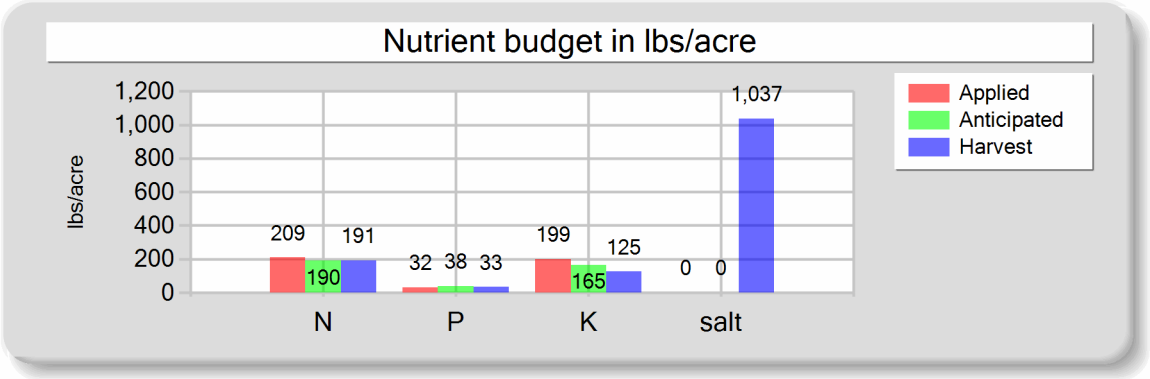
Fresh water applied	
57,618,000.00	*gallons*
2,121.88	*acre-inches*
30.31	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
2	*harvests*

Field #M2 - 06/01/2023: Corn, silage

Field name: Field #M2 Crop: Corn, silage Plant date: 06/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	122.76	31.57	198.75	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	72.15	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	208.91	31.57	198.75	0.00
Anticipated crop nutrient removal	190.00	37.50	165.00	0.00
Actual crop nutrient removal	190.63	33.25	124.59	1,037.00
Nutrient balance	18.28	-1.68	74.15	-1,037.00
Applied to removed ratio	1.10	0.95	1.60	0.00

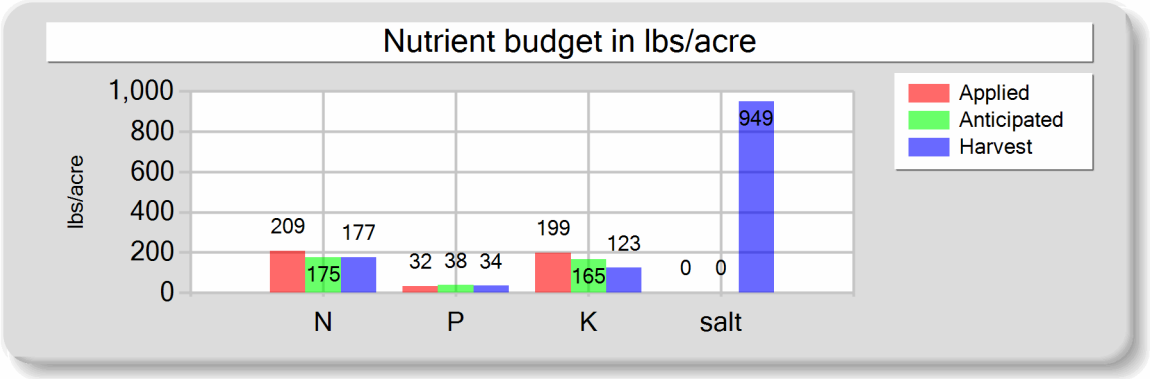
Fresh water applied	
25,938,000.00	*gallons*
955.21	*acre-inches*
31.84	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field #M3 - 06/01/2023: Corn, silage

Field name: Field #M3 Crop: Corn, silage Plant date: 06/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	122.76	31.57	198.75	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	72.15	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	208.91	31.57	198.75	0.00
Anticipated crop nutrient removal	175.00	37.50	165.00	0.00
Actual crop nutrient removal	176.66	34.31	122.93	949.00
Nutrient balance	32.25	-2.74	75.82	-949.00
Applied to removed ratio	1.18	0.92	1.62	0.00

Fresh water applied	
25,938,000.00	*gallons*
955.21	*acre-inches*
31.84	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

T3139072B

Sample and source description: T3139072B

Sample date: 05/19/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 56.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,450.00	2,430.00	15,300.00							0.00
DL	1.00	1.00	1.00							1.00

T3195012A

Sample and source description: T3195012A

Sample date: 07/14/2023 Material type: Scraped material Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 58.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,800.00	3,790.00	23,900.00	16,000.00	6,870.00	5,060.00	3,260.00	1,210.00		49.20
DL	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00		1.00

T3264048B

Sample and source description: T3264048B

Sample date: 09/21/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 11.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,800.00	3,720.00	11,000.00	17,300.00	6,210.00	1,830.00	7,660.00	1,090.00		28.30
DL	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00		1.00

B. PROCESS WASTEWATER ANALYSES

T3059006A

Sample and source description: T3059006A

Sample date: 02/28/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	599.00	398.00	0.00	0.00	54.00	612.00								5,790.00	6,380
DL	1.00	1.00	1.00	1.00	1.00	1.00								100.00	1

T3139072A

Sample and source description: T3139072A

Sample date: 05/19/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	728.00	482.00	0.00	0.00	19.90	210.00								8,510.00	8,780
DL	1.00	1.00	1.00	1.00	1.00	1.00								100.00	1

T3264048A

Sample and source description: T3264048A

Sample date: 09/21/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.70

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	661.00	381.00	0.00	0.00	25.90	989.00	161.00	154.00	293.00	4,140.00	0.00	228.00	244.00	8,710.00	9,860
DL	1.00	1.00	1.00	0.14	1.00	1.00	1.00	1.00	1.00	1.00	2.00	1.00	1.00	100.00	1

C. FRESH WATER ANALYSES

Dairy/Dom. Well

Dairy/Dom. Well

T3354006A

Sample description: T3354006A

Sample date: 12/20/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.73	0.00	0.73								377.00	0
DL	0.50	0.50	0.50								100.00	1

Marengo Irr. Well

T2350004C

Sample description: T2350004C

Sample date: 12/16/2022 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	10.00	0.00	10.00								645.00	0
DL	1.00	0.50	1.00								100.00	1

Tank Well

T3354006B

Sample description: T3354006B

Sample date: 12/20/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1.06	0.00	1.06								365.00	0
DL	1.00	0.50	1.00								100.00	1

D. SOIL ANALYSES

Field #D2

T3354006C

Sample and source description: T3354006C

Sample date: 12/20/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			82.00		0.00	0.00	
DL			1.00		1.00	1.00	

Field #D3

T3354006D

Sample and source description: T3354006D

Sample date: 12/20/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			108.00		0.00	0.00	
DL			1.00		1.00	1.00	

Field #D4

T3354006E

Sample and source description: T3354006E

Sample date: 12/20/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			86.00		0.00	0.00	
DL			1.00		1.00	1.00	

Field #D5

Field #D5

T3354006D

Sample and source description: T3354006D

Sample date: 12/20/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			108.00		0.00	0.00	
DL			1.00		1.00	1.00	

Field #D6

T3354006C

Sample and source description: T3354006C

Sample date: 12/20/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			82.00		0.00	0.00	
DL			1.00		1.00	1.00	

Field #M1

T3354006F

Sample and source description: T3354006F

Sample date: 12/20/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			21.00		0.00	0.00	
DL			1.00		1.00	1.00	

Field #M1

T3354006G

Sample and source description: T3354006G

Sample date: 12/20/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			23.00		0.00	0.00	
DL			1.00		1.00	1.00	

Field #M2

T3354006G

Sample and source description: T3354006G

Sample date: 12/20/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			23.00		0.00	0.00	
DL			1.00		1.00	1.00	

T3354006H

Sample and source description: T3354006H

Sample date: 12/20/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			21.00		0.00	0.00	
DL			1.00		1.00	1.00	

Field #M3

Field #M3

T3354006H

Sample and source description: T3354006H

Sample date: 12/20/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			21.00		0.00	0.00	
DL			1.00		1.00	1.00	

T3354006I

Sample and source description: T3354006I

Sample date: 12/20/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			31.00		0.00	0.00	
DL			1.00		1.00	1.00	

E. PLANT TISSUE ANALYSES

Field #M1 - 06/01/2023: Corn, silage

T3279065A

Sample and source description: T3279065A

Sample date: 10/06/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,100.00	2,120.00	8,340.00		5.30
DL	1.00	1.00	1.00		1.00

Field #M1 - 06/01/2023: Corn, silage

T3279065B

Sample and source description: T3279065B

Sample date: 10/06/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,700.00	2,130.00	7,660.00		6.40
DL	1.00	1.00	1.00		1.00

Field #M2 - 06/01/2023: Corn, silage

T3279065C

Sample and source description: T3279065C

Sample date: 10/06/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,500.00	2,180.00	8,170.00		6.80
DL	1.00	1.00	1.00		1.00

Field #M3 - 06/01/2023: Corn, silage

T3279065D

Sample and source description: T3279065D

Sample date: 10/06/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,100.00	2,350.00	8,420.00		6.50
DL	1.00	1.00	1.00		1.00

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

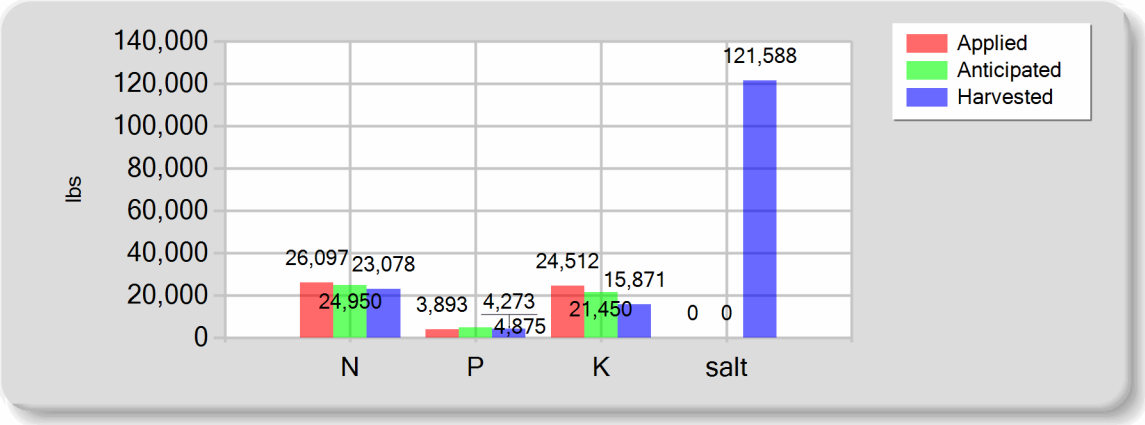
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

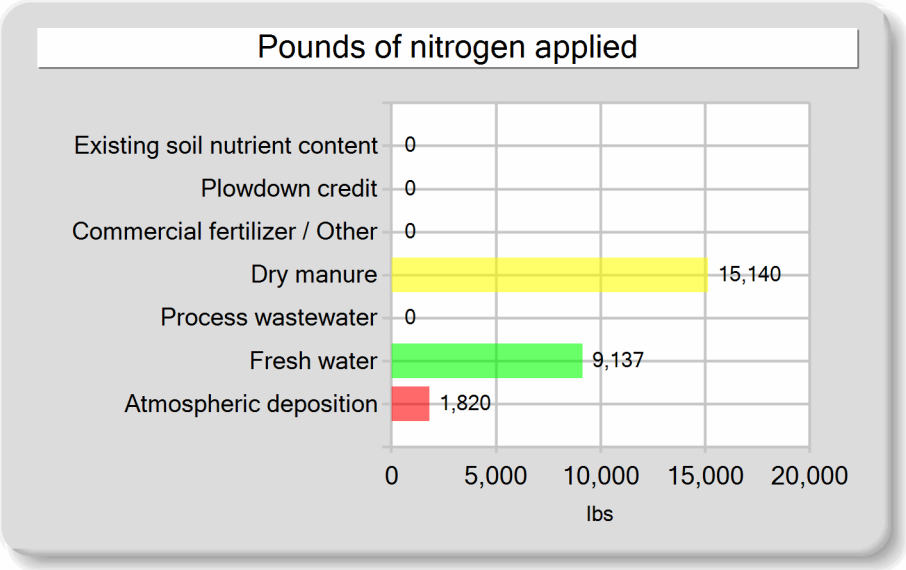
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

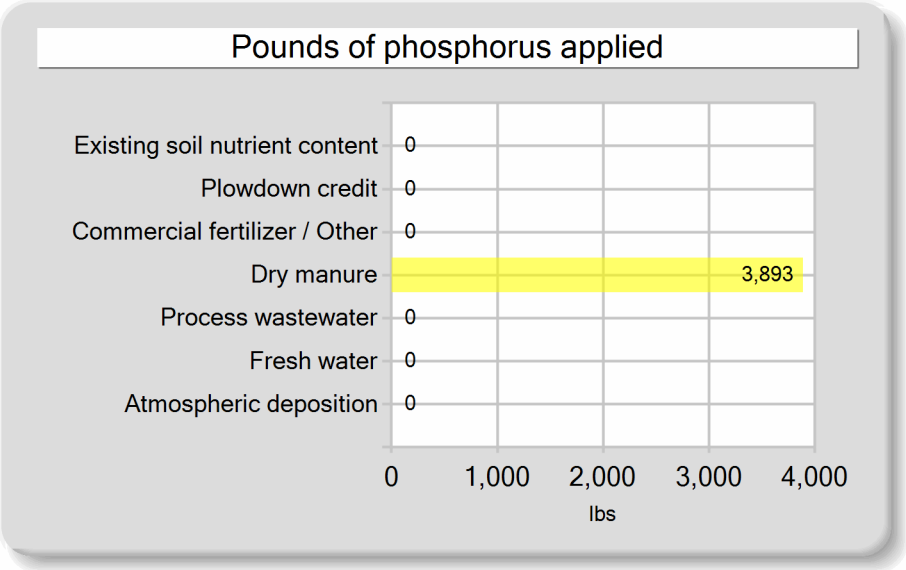
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	15,139.85	3,893.10	24,512.13	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	9,137.27	0.00	0.00	0.00
Atmospheric deposition	1,820.00	0.00	0.00	0.00
Total nutrients applied	26,097.12	3,893.10	24,512.13	0.00
Anticipated crop nutrient removal	24,950.00	4,875.00	21,450.00	0.00
Actual crop nutrient removal	23,077.80	4,272.93	15,871.03	121,587.75
Nutrient balance	3,019.32	-379.83	8,641.11	-121,587.75
Applied to removed ratio	1.13	0.91	1.54	0.00

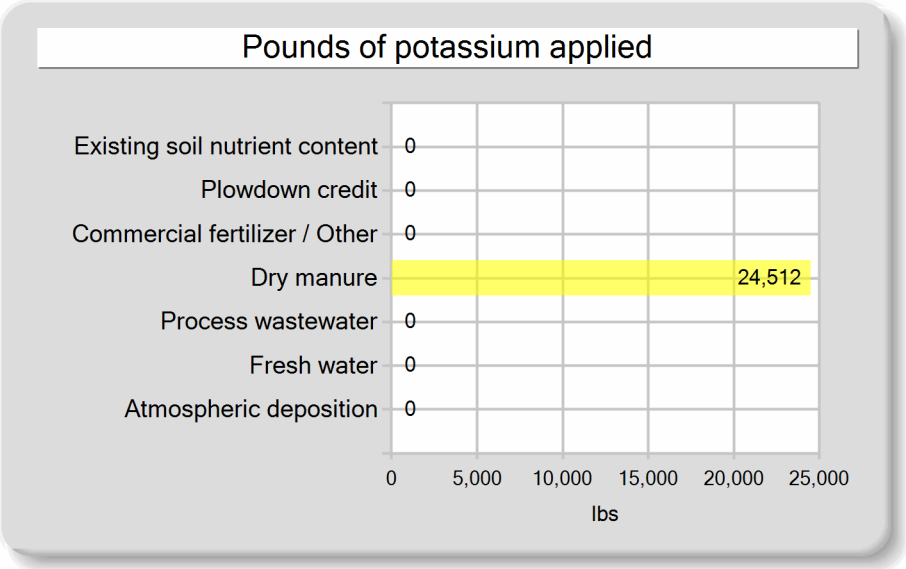
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

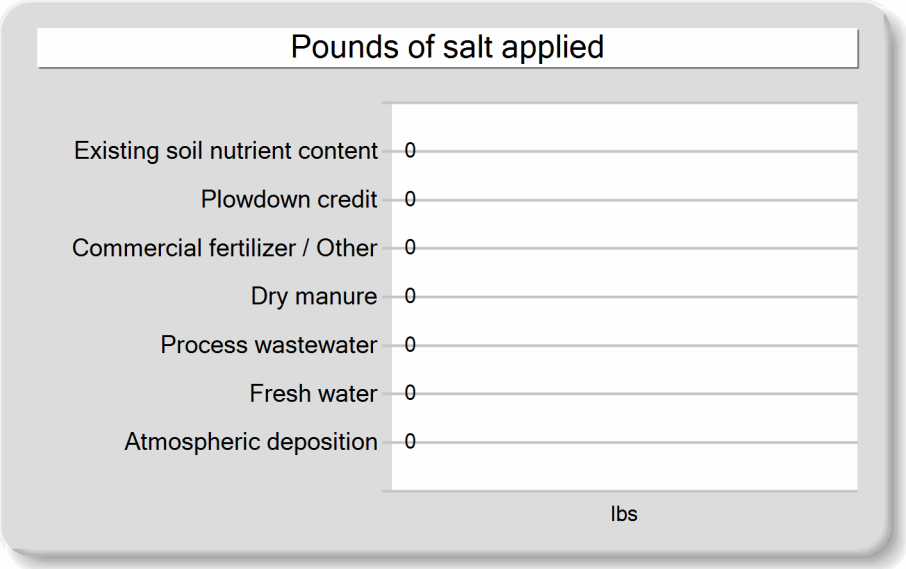


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

Solid manure was only applied for the corn crop on the M Fields during 2023. No wastewater was applied to dairy fields during 2023.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Richard J. Denier	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Cal-Denier Dairy LLC

Physical address of dairy:

Number and Street	City	County	Zip Code
10715 Arno RD	Galt	Sacramento	95632

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 08/01/1981

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X134-X280-X050-XXXX X150-X072-X003-XXXX

B. OPERATORS

Denier, Fred A

Operator name: Denier, Fred A

Telephone no.: (209) 745-4982 Landline, (209) 483-0318 Cellular

Mailing Address Number and Street	City	State	Zip Code
10715 Arno RD	Galt	CA	95632

Denier, Richard J.

Operator name: Denier, Richard J.

Telephone no.: (209) 745-4982 Landline, Cellular

Mailing Address Number and Street	City	State	Zip Code
10715 Arno RD	Galt	CA	95632

This operator is responsible for paying permit fees.

C. OWNERS

Denier, Fred A

Legal owner name: Denier, Fred A

Telephone no.: (209) 745-4982 Landline, (209) 483-0318 Cellular

Mailing Address Number and Street	City	State	Zip Code
10715 Arno RD	Galt	CA	95632

Denier, Richard J.

Legal owner name: Denier, Richard J.

Telephone no.: (209) 745-4982

Landline	Cellular
(209) 745-4982	

Mailing Address Number and Street	City	State	Zip Code
10715 Arno RD	Galt	CA	95632

This owner is responsible for paying permit fees.

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	0	110	0	0
Number under roof	850	100	180	0	0	0
Maximum number	850	100	180	110	0	0
Average number	850	100	180	110	0	0
Avg live weight (lbs)	1,400	1,450	1,000	750		

Predominant milk cow breed: Holstein

Average milk production: 85 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 27,711.51 *tons per reporting period*

Total nitrogen from manure: 350,408.45 *lbs per reporting period*

After ammonia losses (30% loss applied): 245,285.92 *lbs per reporting period*

Total phosphorus from manure: 58,924.91 *lbs per reporting period*

Total potassium from manure: 171,159.59 *lbs per reporting period*

Total salt from manure: 423,217.50 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 3,456,000 *gallons*

Total nitrogen generated: 18,169.40 *lbs*

Total phosphorus generated: 1,152.17 *lbs*

Total potassium generated: 23,086.68 *lbs*

Total salt generated: 234,183.40 *lbs*

	0	*gallons applied*
+	3,456,000	*gallons exported*
-	0	*gallons imported*
=	3,456,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Dairy/Dom. Well	Ground water
Marengo Irr. Well	Ground water
Tank Well	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

Date	Material type	Quantity	Reporting basis	Moisture (%)	Density (lbs/cu ft)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/17/2023	Corral solids	900.00 *ton*	Dry-weight	11.9		20,800.00	3,720.00	11,000.00		28.30

Date	Material type	Quantity	Kjeldahl-N (mg/L)	Ammonium-N (mg/L)	Ammonia-N (mg/L)	Nitrate-N (mg/L)	P (mg/L)	K (mg/L)	EC (µmhos/cm)	TDS (mg/L)
02/28/2023	Process wastewater	1,728,000.00 *gal*	599.00	398.00	0.00	0.00	54.00	612.00		6,380
09/30/2023	Process wastewater	1,728,000.00 *gal*	661.00	381.00	0.00	0.00	25.90	989.00		9,860

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Dry manure	32,984.64	5,899.18	17,443.80	448,781.40
Process wastewater	18,169.40	1,152.17	23,086.68	234,183.40
Total exports for all materials	51,154.04	7,051.35	40,530.48	682,964.80

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Field #D2	35	35	0	none	X134-X280-X050-XXXX
Field #D3	25	25	0	none	X134-X280-X050-XXXX
Field #D4	40	40	0	none	X134-X280-X050-XXXX
Field #D5	15	15	0	none	X134-X280-X050-XXXX
Field #D6	8	8	0	none	X134-X280-X050-XXXX
Field #M1	70	70	2	manure	X150-X072-X003-XXXX
Field #M2	30	30	1	manure	X150-X072-X003-XXXX
Field #M3	30	30	1	manure	X150-X072-X003-XXXX
Totals for areas that were used for application	130	130	4		
Totals for areas that were not used for application	123	123	0		
Land application area totals	253	253	4		

B. CROPS AND HARVESTS

Field #M1

Field name: Field #M1

06/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 70 Plant date: 06/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/03/2023	875.00 *ton*	Dry-weight		70.7	11,100.00	2,120.00	8,340.00		5.30
10/03/2023	875.00 *ton*	Dry-weight		68.9	11,700.00	2,130.00	7,660.00		6.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	25.00	172.28	32.09	120.65	885.83

Field #M2

Field name: Field #M2

Field #M2

06/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 30 Plant date: 06/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/04/2023	750.00 *ton*	Dry-weight		69.5	12,500.00	2,180.00	8,170.00		6.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	190.00	37.50	165.00	0.00
Total actual harvest content	25.00	190.63	33.25	124.59	1,037.00

Field #M3

Field name: Field #M3

06/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 30 Plant date: 06/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/04/2023	750.00 *ton*	Dry-weight		70.8	12,100.00	2,350.00	8,420.00		6.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	175.00	37.50	165.00	0.00
Total actual harvest content	25.00	176.66	34.31	122.93	949.00

NUTRIENT BUDGET

A. LAND APPLICATIONS

Field #M1 - 06/01/2023: Corn, silage

Field name: Field #M1

Crop: Corn, silage

Plant date: 06/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/25/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Application	Corral solids	111.06	28.56	179.82	0.00	950.00 *ton*
Application event totals		111.06	28.56	179.82	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Marengo Irr. Well	Ground water	9.28	0.00	0.00	0.00	7,788,000.00 *gal*
Application event totals		9.28	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Marengo Irr. Well	Ground water	8.89	0.00	0.00	0.00	7,458,000.00 *gal*
Application event totals		8.89	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Marengo Irr. Well	Ground water	8.42	0.00	0.00	0.00	7,062,000.00 *gal*
Application event totals		8.42	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Marengo Irr. Well	Ground water	8.42	0.00	0.00	0.00	7,062,000.00 *gal*
Application event totals		8.42	0.00	0.00	0.00	

Field #M1 - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Marengo Irr. Well	Ground water	8.42	0.00	0.00	0.00	7,062,000.00 *gal*
Application event totals		8.42	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Marengo Irr. Well	Ground water	8.42	0.00	0.00	0.00	7,062,000.00 *gal*
Application event totals		8.42	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Marengo Irr. Well	Ground water	8.42	0.00	0.00	0.00	7,062,000.00 *gal*
Application event totals		8.42	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Marengo Irr. Well	Ground water	8.42	0.00	0.00	0.00	7,062,000.00 *gal*
Application event totals		8.42	0.00	0.00	0.00	

Field #M2 - 06/01/2023: Corn, silage

Field name: Field #M2

Crop: Corn, silage

Plant date: 06/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/25/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Application	Corral solids	122.76	31.57	198.75	0.00	450.00 *ton*
Application event totals		122.76	31.57	198.75	0.00	

Field #M2 - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Marengo Irr. Well	Ground water	10.65	0.00	0.00	0.00	3,828,000.00 *gal*
Application event totals		10.65	0.00	0.00	0.00	

06/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Marengo Irr. Well	Ground water	9.73	0.00	0.00	0.00	3,498,000.00 *gal*
Application event totals		9.73	0.00	0.00	0.00	

07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Marengo Irr. Well	Ground water	8.63	0.00	0.00	0.00	3,102,000.00 *gal*
Application event totals		8.63	0.00	0.00	0.00	

07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Marengo Irr. Well	Ground water	8.63	0.00	0.00	0.00	3,102,000.00 *gal*
Application event totals		8.63	0.00	0.00	0.00	

08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Marengo Irr. Well	Ground water	8.63	0.00	0.00	0.00	3,102,000.00 *gal*
Application event totals		8.63	0.00	0.00	0.00	

08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Marengo Irr. Well	Ground water	8.63	0.00	0.00	0.00	3,102,000.00 *gal*
Application event totals		8.63	0.00	0.00	0.00	

Field #M2 - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Marengo Irr. Well	Ground water	8.63	0.00	0.00	0.00	3,102,000.00 *gal*
Application event totals		8.63	0.00	0.00	0.00	

09/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Marengo Irr. Well	Ground water	8.63	0.00	0.00	0.00	3,102,000.00 *gal*
Application event totals		8.63	0.00	0.00	0.00	

Field #M3 - 06/01/2023: Corn, silage

Field name: Field #M3

Crop: Corn, silage

Plant date: 06/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/25/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Application	Corral solids	122.76	31.57	198.75	0.00	450.00 *ton*
Application event totals		122.76	31.57	198.75	0.00	

06/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Marengo Irr. Well	Ground water	10.65	0.00	0.00	0.00	3,828,000.00 *gal*
Application event totals		10.65	0.00	0.00	0.00	

06/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Marengo Irr. Well	Ground water	9.73	0.00	0.00	0.00	3,498,000.00 *gal*
Application event totals		9.73	0.00	0.00	0.00	

Field #M3 - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Marengo Irr. Well	Ground water	8.63	0.00	0.00	0.00	3,102,000.00 *gal*
Application event totals		8.63	0.00	0.00	0.00	

07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Marengo Irr. Well	Ground water	8.63	0.00	0.00	0.00	3,102,000.00 *gal*
Application event totals		8.63	0.00	0.00	0.00	

08/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Marengo Irr. Well	Ground water	8.63	0.00	0.00	0.00	3,102,000.00 *gal*
Application event totals		8.63	0.00	0.00	0.00	

08/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Marengo Irr. Well	Ground water	8.63	0.00	0.00	0.00	3,102,000.00 *gal*
Application event totals		8.63	0.00	0.00	0.00	

09/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Marengo Irr. Well	Ground water	8.63	0.00	0.00	0.00	3,102,000.00 *gal*
Application event totals		8.63	0.00	0.00	0.00	

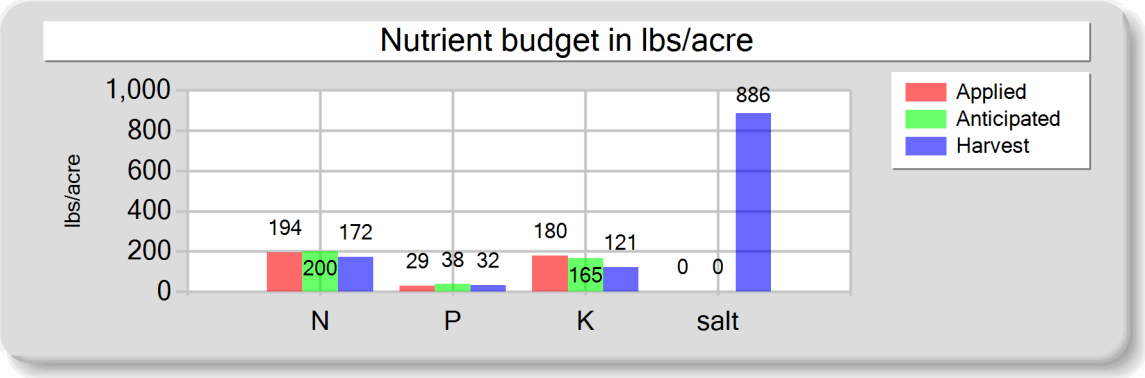
09/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Marengo Irr. Well	Ground water	8.63	0.00	0.00	0.00	3,102,000.00 *gal*
Application event totals		8.63	0.00	0.00	0.00	

B. NUTRIENT BUDGET

Field #M1 - 06/01/2023: Corn, silage

Field name: Field #M1 Crop: Corn, silage Plant date: 06/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	111.06	28.56	179.82	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	68.69	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	193.75	28.56	179.82	0.00
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	172.28	32.09	120.65	885.83
Nutrient balance	21.48	-3.53	59.17	-885.83
Applied to removed ratio	1.12	0.89	1.49	0.00

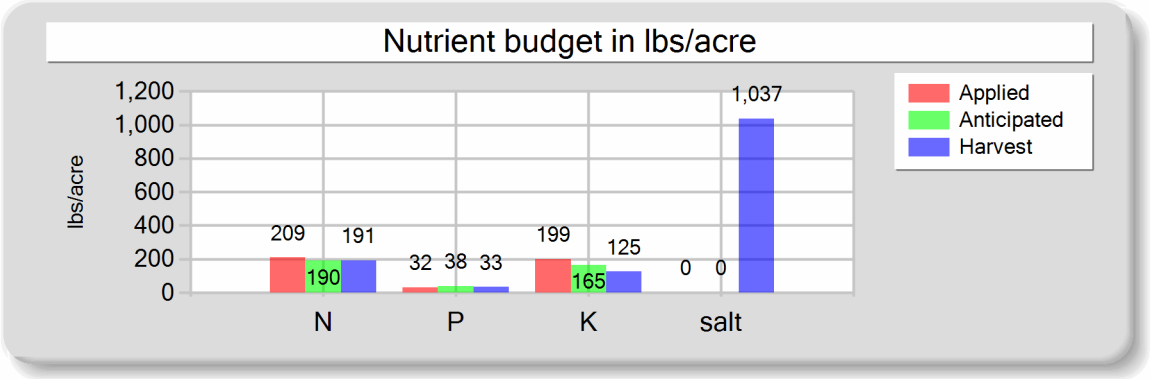
Fresh water applied	
57,618,000.00	*gallons*
2,121.88	*acre-inches*
30.31	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
2	*harvests*

Field #M2 - 06/01/2023: Corn, silage

Field name: Field #M2 Crop: Corn, silage Plant date: 06/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	122.76	31.57	198.75	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	72.15	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	208.91	31.57	198.75	0.00
Anticipated crop nutrient removal	190.00	37.50	165.00	0.00
Actual crop nutrient removal	190.63	33.25	124.59	1,037.00
Nutrient balance	18.28	-1.68	74.15	-1,037.00
Applied to removed ratio	1.10	0.95	1.60	0.00

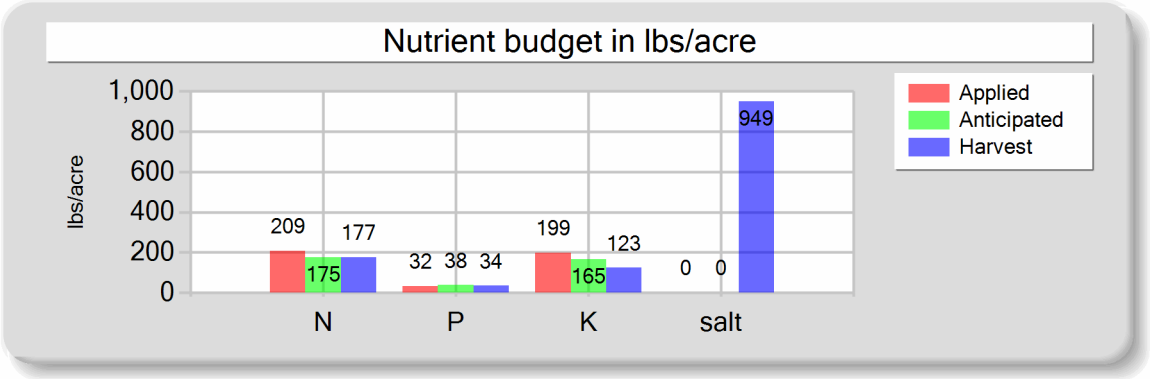
Fresh water applied	
25,938,000.00	*gallons*
955.21	*acre-inches*
31.84	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field #M3 - 06/01/2023: Corn, silage

Field name: Field #M3 Crop: Corn, silage Plant date: 06/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	122.76	31.57	198.75	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	72.15	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	208.91	31.57	198.75	0.00
Anticipated crop nutrient removal	175.00	37.50	165.00	0.00
Actual crop nutrient removal	176.66	34.31	122.93	949.00
Nutrient balance	32.25	-2.74	75.82	-949.00
Applied to removed ratio	1.18	0.92	1.62	0.00

Fresh water applied	
25,938,000.00	*gallons*
955.21	*acre-inches*
31.84	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

T3139072B

Sample and source description: T3139072B

Sample date: 05/19/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 56.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,450.00	2,430.00	15,300.00							0.00
DL	1.00	1.00	1.00							1.00

T3195012A

Sample and source description: T3195012A

Sample date: 07/14/2023 Material type: Scraped material Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 58.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,800.00	3,790.00	23,900.00	16,000.00	6,870.00	5,060.00	3,260.00	1,210.00		49.20
DL	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00		1.00

T3264048B

Sample and source description: T3264048B

Sample date: 09/21/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 11.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,800.00	3,720.00	11,000.00	17,300.00	6,210.00	1,830.00	7,660.00	1,090.00		28.30
DL	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00		1.00

B. PROCESS WASTEWATER ANALYSES

T3059006A

Sample and source description: T3059006A

Sample date: 02/28/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	599.00	398.00	0.00	0.00	54.00	612.00								5,790.00	6,380
DL	1.00	1.00	1.00	1.00	1.00	1.00								100.00	1

T3139072A

Sample and source description: T3139072A

Sample date: 05/19/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	728.00	482.00	0.00	0.00	19.90	210.00								8,510.00	8,780
DL	1.00	1.00	1.00	1.00	1.00	1.00								100.00	1

T3264048A

Sample and source description: T3264048A

Sample date: 09/21/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.70

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	661.00	381.00	0.00	0.00	25.90	989.00	161.00	154.00	293.00	4,140.00	0.00	228.00	244.00	8,710.00	9,860
DL	1.00	1.00	1.00	0.14	1.00	1.00	1.00	1.00	1.00	1.00	2.00	1.00	1.00	100.00	1

C. FRESH WATER ANALYSES

Dairy/Dom. Well

Dairy/Dom. Well

T3354006A

Sample description: T3354006A

Sample date: 12/20/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.73	0.00	0.73								377.00	0
DL	0.50	0.50	0.50								100.00	1

Marengo Irr. Well

T2350004C

Sample description: T2350004C

Sample date: 12/16/2022 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	10.00	0.00	10.00								645.00	0
DL	1.00	0.50	1.00								100.00	1

Tank Well

T3354006B

Sample description: T3354006B

Sample date: 12/20/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1.06	0.00	1.06								365.00	0
DL	1.00	0.50	1.00								100.00	1

D. SOIL ANALYSES

Field #D2

T3354006C

Sample and source description: T3354006C

Sample date: 12/20/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			82.00		0.00	0.00	
DL			1.00		1.00	1.00	

Field #D3

T3354006D

Sample and source description: T3354006D

Sample date: 12/20/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			108.00		0.00	0.00	
DL			1.00		1.00	1.00	

Field #D4

T3354006E

Sample and source description: T3354006E

Sample date: 12/20/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			86.00		0.00	0.00	
DL			1.00		1.00	1.00	

Field #D5

Field #D5

T3354006D

Sample and source description: T3354006D

Sample date: 12/20/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			108.00		0.00	0.00	
DL			1.00		1.00	1.00	

Field #D6

T3354006C

Sample and source description: T3354006C

Sample date: 12/20/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			82.00		0.00	0.00	
DL			1.00		1.00	1.00	

Field #M1

T3354006F

Sample and source description: T3354006F

Sample date: 12/20/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			21.00		0.00	0.00	
DL			1.00		1.00	1.00	

Field #M1

T3354006G

Sample and source description: T3354006G

Sample date: 12/20/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			23.00		0.00	0.00	
DL			1.00		1.00	1.00	

Field #M2

T3354006G

Sample and source description: T3354006G

Sample date: 12/20/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			23.00		0.00	0.00	
DL			1.00		1.00	1.00	

T3354006H

Sample and source description: T3354006H

Sample date: 12/20/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			21.00		0.00	0.00	
DL			1.00		1.00	1.00	

Field #M3

Field #M3

T3354006H

Sample and source description: T3354006H

Sample date: 12/20/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			21.00		0.00	0.00	
DL			1.00		1.00	1.00	

T3354006I

Sample and source description: T3354006I

Sample date: 12/20/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			31.00		0.00	0.00	
DL			1.00		1.00	1.00	

E. PLANT TISSUE ANALYSES

Field #M1 - 06/01/2023: Corn, silage

T3279065A

Sample and source description: T3279065A

Sample date: 10/06/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,100.00	2,120.00	8,340.00		5.30
DL	1.00	1.00	1.00		1.00

Field #M1 - 06/01/2023: Corn, silage

T3279065B

Sample and source description: T3279065B

Sample date: 10/06/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,700.00	2,130.00	7,660.00		6.40
DL	1.00	1.00	1.00		1.00

Field #M2 - 06/01/2023: Corn, silage

T3279065C

Sample and source description: T3279065C

Sample date: 10/06/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,500.00	2,180.00	8,170.00		6.80
DL	1.00	1.00	1.00		1.00

Field #M3 - 06/01/2023: Corn, silage

T3279065D

Sample and source description: T3279065D

Sample date: 10/06/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,100.00	2,350.00	8,420.00		6.50
DL	1.00	1.00	1.00		1.00

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

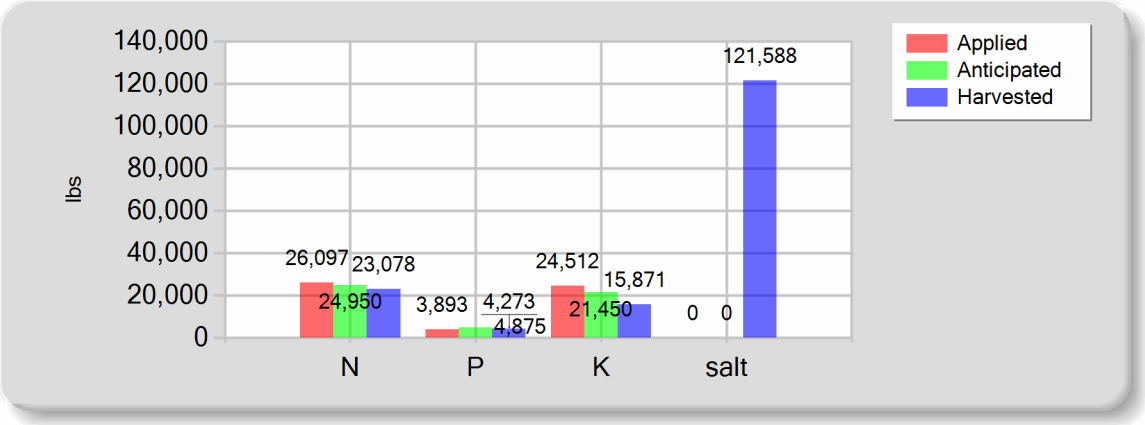
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

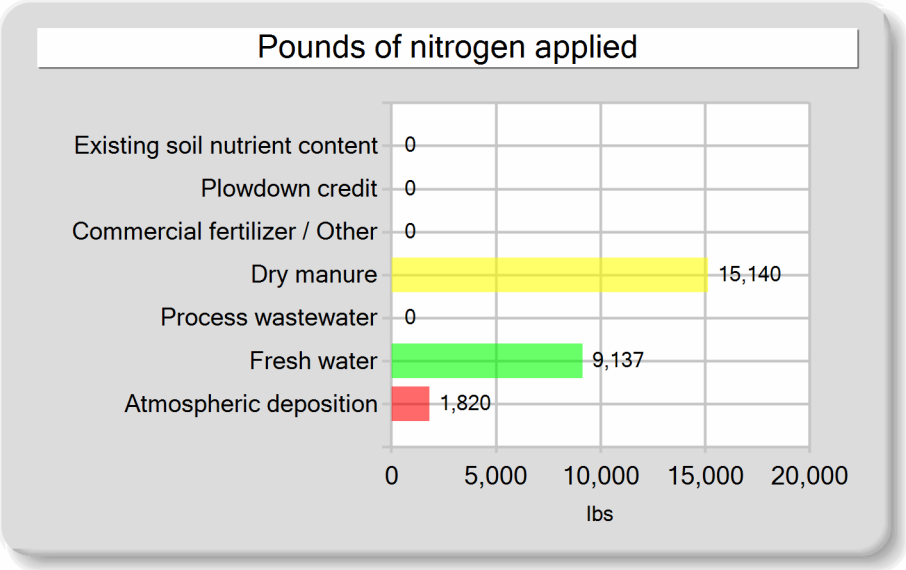
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

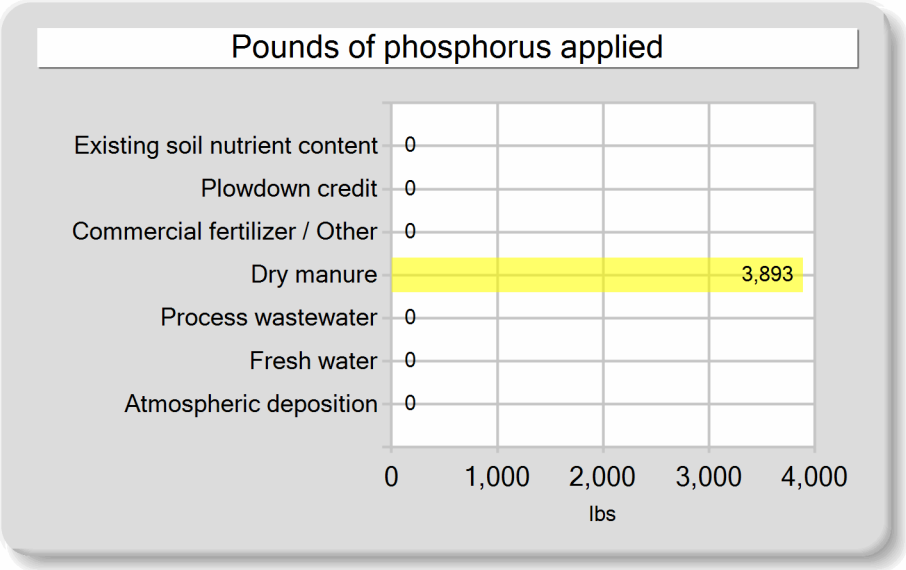
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	15,139.85	3,893.10	24,512.13	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	9,137.27	0.00	0.00	0.00
Atmospheric deposition	1,820.00	0.00	0.00	0.00
Total nutrients applied	26,097.12	3,893.10	24,512.13	0.00
Anticipated crop nutrient removal	24,950.00	4,875.00	21,450.00	0.00
Actual crop nutrient removal	23,077.80	4,272.93	15,871.03	121,587.75
Nutrient balance	3,019.32	-379.83	8,641.11	-121,587.75
Applied to removed ratio	1.13	0.91	1.54	0.00

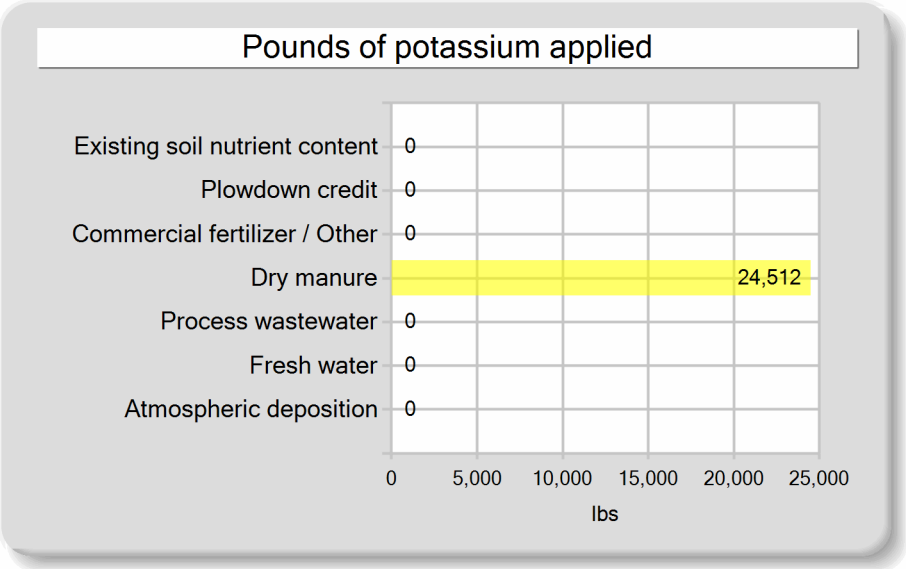
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

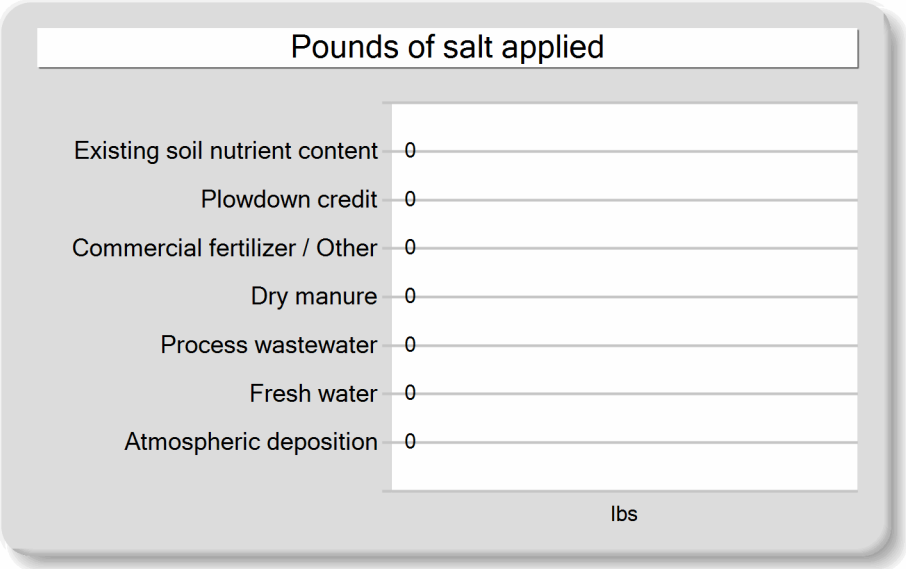


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

Solid manure was only applied for the corn crop on the M Fields during 2023. No wastewater was applied to dairy fields during 2023.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Richard J. Denier
SIGNATURE OF OWNER OF FACILITY

Richard J. Denier
PRINT OR TYPE NAME

6/19/24
DATE

Same as Owner
SIGNATURE OF OPERATOR OF FACILITY

SAME AS OWNER
PRINT OR TYPE NAME

DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination**.
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: ____Richard Denier_____

Name of Dairy Facility: __Cal-Denier Dairy_____

Facility Address: _10715 Arno Rd._____Galt_____95632__
Number and Street City Zip Code

Contact Person Name and Phone Number: __Richard Denier_____831-238-0102__
Name Phone Number

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: _Mark Beck_____

Address of Hauling Company /Person: _3217 W. Forest Lake Rd._____Galt_____95632_
Number and Street City Zip Code

Contact Person: __Mark Beck_____209-610-2855_____
Name Phone Number

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) _____(please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

__Salas Ranch_____10630 Salas Rd_____Wilton_____95639___209-610-2855__
Name Number and Street City Zip Code Phone Number

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

_____X134-0250-X001-0000 & X134-0230-X009-0000_____
Number and Street City Zip Code Assessor's Parcel Number

Dates Hauled: ____10/17/23_____

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount: N 2.08%
P 0.372% K 1.10%

Manure: ___900_____Tons or Cubic Yards (indicate which units used)
Manure Solids Content (if amount reported in tons): ___11.9%_____
Manure Density (if amount reported in cubic yards): _____

Method used to determine amount of manure: _____

_____

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: _____ Gallons

Method used to determine volume of process wastewater: _____

_____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

____ Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: Richard Denier Date: 3/20/24

Hauler's Signature: [illegible] Date: 3/20/24

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination**.
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: ____Richard Denier_____

Name of Dairy Facility: __Cal-Denier Dairy_____

Facility Address: _10715 Arno Rd._____Galt_____95632__
Number and Street City Zip Code

Contact Person Name and Phone Number: __Richard Denier_____831-238-0102__
Name Phone Number

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: _Pipeline from Dairy_____

Address of Hauling Company /Person: _10715 Arno Rd._____Galt_____95632_
Number and Street City Zip Code

Contact Person: __Richard Denier_____831-238-0102_____
Name Phone Number

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) _____(please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

_B & J Farms_____10490 Arno Rd._____Galt_____95632____209-346-6173_
Name Number and Street City Zip Code Phone Number

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

_____X134-0280-X034-0000_
Number and Street City Zip Code Assessor's Parcel Number

Dates Hauled: ____February 2023_____

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount: TKN 599 mg/L
TP 54.0 mg/L TK 612 mg/L

Manure: _____Tons or Cubic Yards (indicate which units used)
Manure Solids Content (if amount reported in tons): _____
Manure Density (if amount reported in cubic yards): _____

Method used to determine amount of manure: _____

_____

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: 1,728,000 Gallons

Method used to determine volume of process wastewater: 400 gpm x time

_____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

__X__ Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: Rod Oemen Date: 12/5/23

Hauler's Signature: Rod Oemen Date: 12/5/23

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination**.
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: ____Richard Denier_____

Name of Dairy Facility: __Cal-Denier Dairy_____

Facility Address: _10715 Arno Rd._____Galt_____95632__
Number and Street City Zip Code

Contact Person Name and Phone Number: __Richard Denier_____831-238-0102__
Name Phone Number

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: _Pipeline from Dairy_____

Address of Hauling Company /Person: _10715 Arno Rd._____Galt_____95632_
Number and Street City Zip Code

Contact Person: __Richard Denier_____831-238-0102_____
Name Phone Number

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) _____(please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

_B & J Farms_____10490 Arno Rd._____Galt_____95632____209-346-6173_
Name Number and Street City Zip Code Phone Number

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

_____X134-0280-X034-0000_
Number and Street City Zip Code Assessor's Parcel Number

Dates Hauled: ____July-September 2023_____

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount: TKN 661 mg/L
TP 25.9 mg/L TK 989 mg/L

Manure: _____Tons or Cubic Yards (indicate which units used)
Manure Solids Content (if amount reported in tons): _____
Manure Density (if amount reported in cubic yards): _____

Method used to determine amount of manure: _____

_____

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: 1,728,000 Gallons

Method used to determine volume of process wastewater: 400 gpm X time

_____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

__X__ Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: Fred Oamen Date: 12/5/23

Hauler's Signature: Fred Oamen Date: 12/5/23

DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380

PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: CAL-DENIER DAIRY
Mailing address: 10715 ARNO RD.
(not facility address) GALT, CA 95632
Phone#: (209) 483-0318

Authorized Copy Released To: JCRK, INC.
(209) 327-0992
Sampled By: JIM CAPTEIN
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
H1 EC*, NO3-N, NH4-N*
H6 EC, TN, TDS
H7 Combination of H1 & H6
H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
H3 EC*, NH4-N, TKN, TP, K, TDS
H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
H8 Combination of H3 &H5
H9 Combination of H4 &H5

TILE DRAIN
H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*
___ (Check here if performed)

DISCHARGE WATER
E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
M1 % Moisture
M3 TN TP, K, % Moisture
M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
M7 Combination of M3 & M4

PLANT TISSUE
F18 % Moisture, Ash, TN, P, K, DM and as received
F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
K3 Soluble Phosphorus
K4 1ft NO3, Organic Matter
K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
K7 Combination of K3 & K4
K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
1	12/18/23	1:00p	Dairy/Dom Well	Dom. Well	15.0'c	20 min	X				
2	✓	1:30p	Tank Well	✓	15.2'c	✓	X				
3	✓	2:00p	Fields D2 + D6	Soil							K3
4	✓	2:15p	Fields D3 + D5	✓							K3
5	✓	2:30p	Field D4	✓							K3
6	✓	3:00p	Field M1	✓							K3
7	✓	3:15p	Fields M1 + M2	✓							K3
8	✓	3:30p	Fields M2 + M3	✓							K3
9	✓	3:45p	Field M3	✓							K3
10											

Analysis

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	(LABORATORY USE ONLY) Logged In By:	LABORATORY NUMBER
1st	Jacquie Captein	JCRK, INC.		12/20/23 8:00a		
2nd		DENELE ANALYTICAL, INC.	12-20-23 12:30			T3354006

Required to be submitted to any agency? (YES / NO)

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
Cal-Denier Dairy, LLC.
10715 Arno Rd.
Galt, CA 95632

Grower

Consultant:
Jacquie Captein
P.O. Box 386
Herald, CA 95638

PURCHASE ORDER:
RECEIVED DATE: 12/20/23 12:30 PM
SUBMITTED BY: Courier
ANALYZED DATE: 12/21/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 15.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Dairy/Dom. Well	T3354006A	Electrical Conductivity (EC)	EPA 120.1 and	0.377	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	0.734	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	242	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	91.4	92.8 (T3355001A)	94.6 (T3355001A)	1.92
Electrical Conductivity (EC)	99.8			0.434
Nitrate Nitrogen (NO3-N)	104	12.7 (T3354012A)	13.9 (T3354012A)	8.73

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

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Compliance Analysis Report

Customer
Cal-Denier Dairy, LLC.
10715 Arno Rd.
Galt, CA 95632

Grower

Consultant:
Jacquie Captein
P.O. Box 386
Herald, CA 95638

PURCHASE ORDER:
RECEIVED DATE: 12/20/23 12:30 PM
SUBMITTED BY: Courier
ANALYZED DATE: 12/21/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 15.2c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Tank Well	T3354006B	Electrical Conductivity (EC)	EPA 120.1 and	0.365	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	1.06	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	234	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	91.4	92.8 (T3355001A)	94.6 (T3355001A)	1.92
Electrical Conductivity (EC)	99.8			0.434
Nitrate Nitrogen (NO3-N)	104	12.7 (T3354012A)	13.9 (T3354012A)	8.73

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ELAP Certificate No.: 2714

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Compliance Analysis Report

Customer
Cal-Denier Dairy
10715 Arno Rd.
Galt, CA 95632

Grower

Consultant:
Jacquie Captein
P.O. Box 386
Herald, CA 95638

PURCHASE ORDER:
RECEIVED DATE: 12/16/22 11:00 AM
SUBMITTED BY: Jacquie
ANALYZED DATE: 12/29/22

Irrigation/ Ground Water (H1)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 10.8c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Marengo Irr. Well	T2350004C	Electrical Conductivity (EC)	EPA 120.1 and	0.645	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	10.0	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	413	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	91.0			
Electrical Conductivity (EC)	100			2.47
Nitrate Nitrogen (NO3-N)	98.6	107 (T2353002D)	106 (T2353002D)	1.37

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ELAP Certificate No.: 2714